OPTIMIZING SATELLITE PLACEMENT WITH *QAOA*



Fig1: Current mega constellation of LEO satellites (Image Source: https://satellitetracker3d.com/)

SATELLITE CHAOS IN SPACE ...

As of April 2025, over **7,200 satellites** orbit in Low-Earth-Orbit (LEO) and the number is expected to reach **100,000** in the next years.[1]

Why?

Companies like Starlink are launching massive constellations to deliver fast internet.
But this brings to:
– **Space traffic & debris**
– **Environmental risks**

Satellite placement is key, but it's an **NP-Hard** problem (not solvable in polynomial time) with many variables:
🌍 Earth coverage
🌀 Orbital mechanics
🧱 Space Debris

Solution?

We need Hybrid Quantum-Classical Algorithms like the **Quantum Approximate Optimization Algorithm (QAOA)**

WHAT IS QAOA?

The Quantum Approximate Optimization Algorithm (QAOA) is a method for solving combinatorial optimization problems.[2] QAOA uses quantum gates to **explore solutions** in parallel and refine them over time.

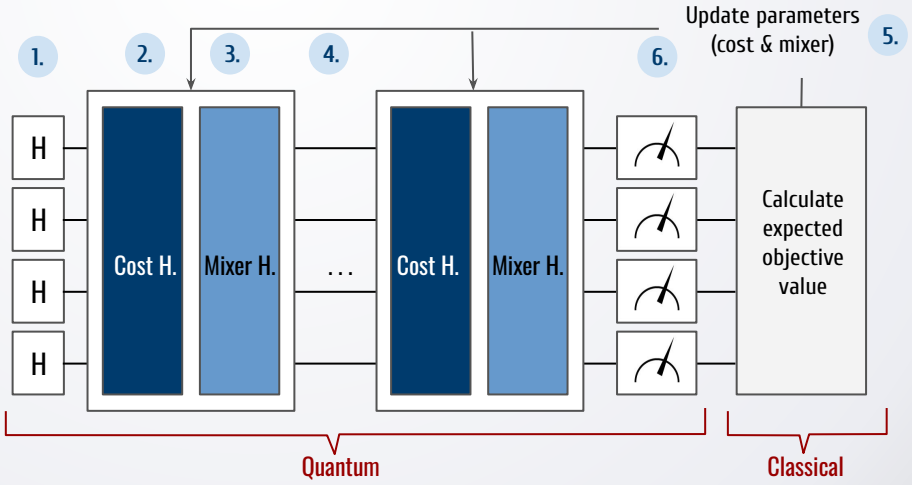
WHY IS QAOA IMPORTANT?

Classical computers struggle with optimization problems that grow exponentially in complexity, like scheduling, logistics, and satellite placement. QAOA can potentially solve these problems much **faster** and more **efficiently.**

HOW DOES QAOA WORK?

The algorithm works by encoding an optimization problem into two Hamiltonians. The steps are:

1. 🎯**Initialization:** We put qubits in superposition (H gates).
2. 🧠 **Cost Hamiltonian:** Encodes the cost function, giving high energy to bad solutions.
3. 🔀**Mixing Hamiltonian:** Explores possible configurations by guiding quantum transitions.
4. 🔛**Repetition:** Iterations of cost (H_C)and mix (H_M)hamiltonians
5. 💻 **Classical Optimization:** A classical computer updates the hamiltonian parameters.
6. 📏 **Measurement:** We finally get the near-optimal solution!



WHY USE QAOA FOR SATELLITES? QAOA optimizes satellite placement by finding the best configuration through a quantum system, reducing overlap and improving coverage and communication.

1. Jajodia, Babita & Lalwani, Jitesh & Bach, B & Patel, Amit & P, A. (2022). White Paper: *Satellite Placement Optimization Using Quantum Computing*.
2. Ceroni, J. (2020, November 17). *Intro to QAOA*. Pennylane. https://pennylane.ai/qml/demos/tutorial_qaoa_intro

QUBIT X QUBIT

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